F. Entitlements, Plans & Commercial Controls

Standalone capability brief · Suggested module: **Entitlements + Usage Metering Platform**

Purpose

Give product domains a single, reliable way to evaluate plan capabilities, quotas, trials, subscription states, and billable usage without duplicating commercial policy.

1. Scope and capability areas

Plans and entitlements

Represent plans, feature capabilities, limits, add-ons, and customer-specific grants.

Subscription lifecycle

Model trial, active, grace, suspended, canceled, and grandfathered states with explicit transitions.

Usage metering

Collect billable events with stable schemas, timestamps, units, source identity, and deduplication keys.

Limits and overages

Support soft warnings, hard enforcement, burst allowances, overage policies, and quota windows.

Caching and invalidation

Define safe caching for entitlement decisions and reliable invalidation when commercial state changes.

Migration

Support plan changes, grandfathered customers, staged migrations, and rollback or reconciliation procedures.

2. Shared contracts and standards

- Expose a consistent entitlement-check API with explicit subject, resource, capability, and evaluation time.
- Distinguish product authorization from commercial entitlement: a user may be authorized but not entitled to a feature.
- Define whether each quota is advisory, soft-limited, or hard-enforced and where enforcement occurs.
- Usage events are idempotent and auditable; billing calculations must be reconcilable to source events.

3. Recommended implementation approach

- Create a versioned entitlement model and a central administrative workflow for plan changes.
- Use local caches only with documented TTLs, invalidation, and safe behavior during control-plane outages.
- Emit usage events at the authoritative point of consumption, not from unreliable UI interactions.
- Build reconciliation jobs to compare source usage, metered usage, and billed quantities.
- Provide sandbox plans and test fixtures for trial, upgrade, downgrade, suspension, and overage scenarios.

4. Failure behavior and operational expectations

- If entitlement evaluation is unavailable, choose behavior by capability risk: preserve safe reads where appropriate but do not grant expensive or restricted capabilities by default.
- Prevent duplicate metering through idempotency keys and replay-safe ingestion.
- Handle stale caches explicitly and expose the decision age for debugging.

5. Suggested operational metrics

- Entitlement decision latency and error rate
- Quota enforcement correctness
- Metering-to-billing reconciliation variance
- Duplicate or missing usage events
- Stale entitlement cache duration

6. Readiness and acceptance checklist

- ■ Product modules use one entitlement interface.
- ■ Plan transitions have deterministic behavior.
- ■ Usage events can be replayed and reconciled.
- ■ Quota semantics are documented and tested.
- ■ Outage behavior is defined per capability risk.

Ownership note: assign a named team, service owner, support route, versioning policy, and deprecation process before treating this capability as a platform contract.